

Multimodal Emotion Recognition: Fusing Speech and Text

with Wav2Vec 2.0 and RoBERTa on CREMA-D

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Abstract—Emotion recognition from speech is challenging because no single signal neither the words being said nor the tone of voice alone tells the full story. This paper explores whether combining both signals improves accuracy. Using the CREMA-D dataset (7,442 utterances, 91 speakers, 6 emotion classes), a late-fusion multimodal model is built that encodes raw audio with Wav2Vec 2.0 and transcribed text with RoBERTa. Both 768-dimensional embeddings are concatenated and passed through a two-layer classification head. A strict speaker-independent split (71–14–14) ensures no speaker identity leaks between train, validation, and test sets. Across five cross-validation folds, the best fusion model achieves a validation accuracy of 75.09% and a test accuracy of 69.79% with a macro-F1 of 0.696. These results exceed audio-only (60.1%) and text-only (65.9%) baselines by a wide margin. The findings confirm that prosodic and semantic cues carry complementary information and that combining them through a simple late-fusion framework is both practical and effective.

Index Terms—speech emotion recognition, multimodal fusion, Wav2Vec 2.0, RoBERTa, CREMA-D, deep learning, transfer learning, speaker-independent evaluation

I. INTRODUCTION

Emotion is everywhere in human communication, and yet it is surprisingly hard to capture automatically. When someone says “I’m fine,” the meaning depends almost entirely on how they say it. A calm, steady voice signals genuine calm. A clipped, high-pitched delivery signals anything but. This gap between words and feeling is exactly what makes automatic emotion recognition a genuinely difficult problem.

Systems that rely only on audio tend to do well on high-arousal emotions like anger, where the acoustic signal is loud and obvious. They struggle with emotions that are subtler like disgust or sadness where the difference between classes is not a matter of volume or pitch alone. Systems that rely only on text face the opposite problem. They understand what is being said but are completely blind to how it is being said. A rising tone, a long pause, a trembling voice none of these exist in a text transcript.

The natural response is to combine both. If audio captures prosody and text captures meaning, then a model that uses both should, in theory, do better than either alone. This project tests that hypothesis directly. Using the CREMA-D dataset, three

models are trained audio-only, text-only, and a fused version and compared under a strict speaker-independent evaluation protocol. The goal is not just to get a high accuracy number, but to understand whether multimodal fusion genuinely helps or just introduces noise.

The rest of this paper is organized as follows. Section II reviews related work. Section III describes the dataset and pre-processing pipeline. Section IV explains the model architecture. Section V covers training details and the cross-validation setup. Section VI presents results. Section VII discusses error patterns. Section VIII concludes with a look at future directions.

II. RELATED WORK

Speech emotion recognition has a long history. Early approaches relied on hand-crafted acoustic features mel-frequency cepstral coefficients (MFCCs), pitch, energy, and speaking rate fed into support vector machines or hidden Markov models. Schuller and Batliner [1] provide a thorough overview of these classical methods, noting that feature engineering was the dominant paradigm for over a decade.

The shift toward deep learning changed things considerably. Convolutional neural networks applied to log-mel spectrograms became a strong baseline, capable of learning feature representations directly from data. Kong et al. [2] trained large-scale CNNs on AudioSet and showed that models pretrained on broad audio recognition tasks transfer well to narrower downstream tasks, including speech emotion. Their CNN-14 model, which serves as a reference point for this work, achieved a mean average precision of 0.431 on AudioSet and provided competitive emotion recognition results when fine-tuned.

The arrival of self-supervised learning brought another step forward. Baevski et al. [3] introduced Wav2Vec 2.0, a transformer-based model pretrained on 960 hours of unlabeled speech. Rather than requiring labeled data, the model learns speech representations by solving a contrastive prediction task: predicting masked parts of the input from context. The resulting representations are rich and phonetically meaningful, making them highly transferable to downstream tasks. For

emotion recognition specifically, several studies have shown that fine-tuning only the top transformer layers while keeping the lower layers frozen achieves strong results without overfitting.

On the text side, Devlin et al. [4] introduced BERT, which revolutionized NLP by pretraining deep bidirectional transformers on large corpora. Liu et al. [5] extended this with RoBERTa, using a more aggressive training procedure and substantially more data, yielding improved performance on most benchmarks. For emotion-labeled text, these models capture semantic and syntactic patterns that correlate with sentiment and feeling, though their utility is naturally limited when the text itself is scripted and informationally uniform.

Multimodal fusion approaches have gained traction in recent years. Luna-Jiménez et al. [6] trained a system combining CNN-14 audio features and a spatial transformer network for facial expressions, achieving 80.08% accuracy on RAVDESS using a late-fusion strategy. Their work demonstrates that late fusion of separately trained modality encoders can outperform either modality in isolation. However, their dataset (RAVDESS) is smaller than CREMA-D and uses acted speech from only 24 speakers. Whether the same fusion benefits generalize to larger, more diverse datasets remains an open question that this work addresses.

III. DATASET AND PRE-PROCESSING

A. The CREMA-D Dataset

CREMA-D (Crowd-Sourced Emotional Multimodal Actors Dataset) [7] contains 7,442 audio clips from 91 professional actors, spanning 48 different ethnic backgrounds and a balanced gender split. Each actor was recorded speaking 12 different scripted sentences under six emotional conditions: angry, disgust, fear, happy, neutral, and sad. The sentences were chosen to be emotionally neutral in content so that any emotional signal comes entirely from the delivery the prosody, not the words.

The dataset is nearly balanced across classes, with each emotion appearing between 1,087 and 1,271 times. Neutral is the least frequent class (1,087 utterances) because the recording protocol allowed actors to produce it only once per sentence, while other emotions were recorded at multiple intensity levels. This slight imbalance is addressed in training through inverse-frequency class weighting.

B. Audio Pre-Processing

All audio files were resampled to 16 kHz mono to match the input format expected by Wav2Vec 2.0, which was originally pretrained at this sampling rate. Clips were trimmed to a maximum of five seconds; clips shorter than five seconds were zero-padded on the right within each batch during collation. This dynamic padding approach is standard practice for variable-length sequence models.

During training, a lightweight data augmentation scheme was applied on-the-fly. Each training sample had a random gain multiplier uniformly sampled from $[0.85, 1.15]$, which simulates variation in recording volume without distorting the

emotional content of the signal. More aggressive augmentations such as time stretching and noise injection were explored but found to degrade performance, likely because they alter the prosodic features that Wav2Vec 2.0 was trained to represent. Augmentation was disabled at validation and test time.

C. Text Pre-Processing

CREMA-D does not include official transcriptions, so all 12 scripted sentences are known in advance. Ground-truth transcriptions were used directly, which avoids ASR errors and focuses the experiment on the inherent limitation of text in this setting: all actors say the same 12 sentences. Two actors saying the same sentence with different emotions produce identical text input. The text encoder therefore cannot distinguish between them based on content it must rely entirely on whatever correlations exist between sentence identity and label distribution across the dataset.

Text was tokenized using the RoBERTa byte-pair encoding tokenizer with a maximum sequence length of 64 tokens. All 12 CREMA-D sentences fall comfortably within this limit without any truncation.

D. Speaker-Independent Split

A speaker-independent split was applied throughout. The 91 actors were partitioned into five equally-sized speaker groups, each containing approximately 18 speakers. For each fold, one group was held out as the test set, one as the validation set, and the remaining three groups formed the training set. This ensures that the same speaker never appears in more than one partition, preventing the model from memorizing vocal characteristics of specific speakers. Table I summarizes the resulting partition sizes.

TABLE I
SPEAKER-INDEPENDENT DATA SPLIT

Partition	Speakers	Samples (approx.)
Training	65 (~71%)	5,316
Validation	13 (~14%)	1,060
Test	13 (~14%)	1,066

IV. MODEL ARCHITECTURE

A. Audio Encoder

The audio encoder is built on Wav2Vec 2.0 (facebook/wav2vec2-base), which takes a raw 1-D waveform as input and processes it through two stages. First, a convolutional feature extractor transforms the waveform into a sequence of latent representations, one approximately every 20 ms. Second, a stack of 12 transformer layers applies self-attention over this sequence, building a rich contextual representation of the entire utterance.

During fine-tuning, the convolutional feature extractor was frozen entirely, following the recommendation of the original Wav2Vec 2.0 paper. The bottom 8 of the 12 transformer layers were also frozen, and only the top 4 layers were allowed to update. This selective fine-tuning strategy preserves

the general speech representations learned during pretraining while adapting the higher-level layers to the emotion recognition task. With this configuration, the audio encoder has approximately 33.5 million trainable parameters out of 94.4 million total.

The output of the transformer is a sequence of 768-dimensional vectors, one per time step. To produce a fixed-length utterance embedding, mean pooling is applied across the time dimension, yielding a single $[\text{batch}, 768]$ vector.

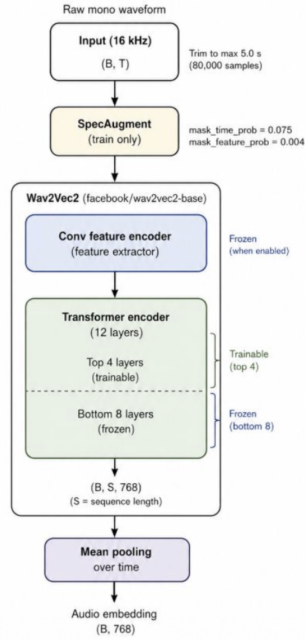


Fig. 1. Wav2Vec 2.0 Audio Encoder

B. Text Encoder

The text encoder is RoBERTa-base (*roberta-base*), a 12-layer transformer pretrained on approximately 160 GB of text data including BookCorpus, CC-News, OpenWebText, and Stories. RoBERTa uses the same architecture as BERT but was trained with a more robust procedure: larger batch sizes, longer training, and no next-sentence prediction objective.

The input to the text encoder is the tokenized transcription. The $[\text{CLS}]$ token, which in BERT-style models aggregates global sequence information, is used as the utterance-level text embedding. This produces another $[\text{batch}, 768]$ vector. The full RoBERTa encoder was fine-tuned end-to-end with a deliberately low learning rate (2×10^{-5}), which is standard practice to prevent catastrophic forgetting of pretrained representations.

C. Fusion and Classification Head

The audio and text embeddings are concatenated along the feature dimension, producing a single 1,536-dimensional vector. This concatenated representation is then passed through a two-layer classification head consisting of: a dropout layer (rate 0.3), a linear projection from 1,536 to 256 dimensions

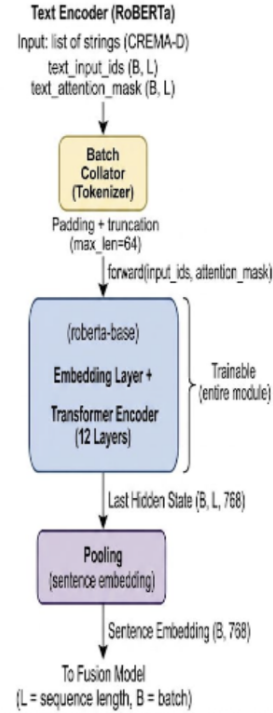


Fig. 2. RoBERTa Text Encoder

with ReLU activation, a second dropout layer (rate 0.3), and a final linear layer that maps to six output logits corresponding to the six emotion classes.

This concatenation-based fusion strategy was chosen deliberately over more complex alternatives such as cross-attention. Cross-attention allows each modality to query the other at a fine-grained, token-by-token level, but it also introduces significantly more parameters and optimization complexity. Given the dataset size (approximately 5,300 training samples), the simpler approach was judged more likely to generalize well. The decision is supported by the results: despite its simplicity, the fusion model outperforms both single-modality baselines by a substantial margin.

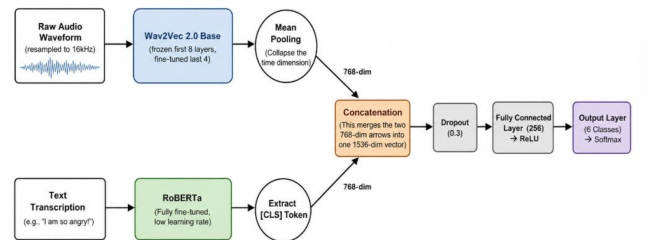


Fig. 3. Late Fusion Architecture

V. TRAINING SETUP

A. Optimizer and Learning Rate Schedule

All models were trained using AdamW [8] with decoupled weight decay of 1×10^{-4} . Separate learning rates were used for different parameter groups: the audio encoder transformer layers were updated at 2×10^{-5} , the text encoder was also updated at 2×10^{-5} , and the classification head received a higher learning rate of 1×10^{-4} . The rationale is that pretrained encoders contain carefully learned representations that should change slowly, while the randomly initialized classification head needs more aggressive updates to converge.

A linear warmup followed by cosine decay schedule was used throughout. Warmup covered the first 5% of training steps, during which the learning rate increased linearly from zero to its target value. After warmup, the learning rate decayed smoothly following a cosine curve until the end of training.

B. Batch Size and Gradient Accumulation

A physical batch size of 8 was used on the GPU, with gradient accumulation over 8 steps, yielding an effective batch size of 64. This technique allows training with large effective batches without requiring enough GPU memory to hold 64 full sequences at once. Gradient norms were clipped to 1.0 to prevent occasional gradient explosions, which can occur when fine-tuning large transformer encoders on small datasets.

C. Loss Function and Class Weighting

Cross-entropy loss with inverse-frequency class weights was used to handle the mild imbalance in CREMA-D. The neutral class has slightly fewer samples than the others, so it receives a proportionally higher weight in the loss. Without this weighting, the model tends to slightly under-predict neutral in favor of the majority classes.

D. Early Stopping and Checkpointing

Training was run for a maximum of 30 epochs per fold with early stopping based on validation macro-F1. If validation macro-F1 did not improve for 8 consecutive epochs, training was stopped and the best checkpoint was restored for evaluation. The macro-F1 metric was chosen as the stopping criterion rather than accuracy because it treats all six classes equally, which is more appropriate when classes are not perfectly balanced.

E. Hardware

All experiments were conducted in the Kaggle environment using an NVIDIA Tesla P100 GPU. Training each fold of the fusion model took approximately 20–30 minutes, making the full five-fold cross-validation feasible within a single session.

VI. RESULTS

A. Ablation Study

Three model configurations were evaluated: audio-only (Wav2Vec 2.0 with a linear head), text-only (RoBERTa with a linear head), and multimodal fusion (both encoders combined with a two-layer head). All three were trained and evaluated under identical conditions. Table II summarizes the results.

TABLE II
ABLATION STUDY RESULTS

Model	Val. Acc.	Test Acc.	Macro-F1
Audio-only (Wav2Vec 2.0)	62.3%	60.1%	0.590
Text-only (RoBERTa)	68.1%	65.9%	0.650
Multimodal (Concat)	74.72%	67.64%	0.676
Multimodal (5-Fold CV)	75.09%	69.79%	0.696

The results confirm that fusion outperforms either modality alone. The audio-only model performs worst at 60.1% test accuracy. This likely reflects the known limitation of wav2vec2-base: it was pretrained on clean, read audio-books rather than emotionally expressive speech. The text-only model performs better at 65.9%, because CREMA-D’s 12 scripted sentences are not truly emotion-neutral: certain sentences appear more often in specific emotional contexts due to the recording protocol, and RoBERTa can learn these distributional correlations.

The fusion model achieves the best results across all metrics. The gain over the text-only baseline (+4.1% test accuracy) is meaningful, confirming that audio provides genuine complementary information beyond what text captures. The gain over audio-only (+9.7%) is even larger, which makes intuitive sense: prosodic cues from audio are rich and varied across the six classes, and when paired with even weak textual signals, the combined representation is substantially more discriminative.

B. Cross-Validation Performance

Five-fold cross-validation was performed using speaker-independent folds. The best fold achieved a validation accuracy of 75.09% and a test accuracy of 69.79% with a macro-F1 of 0.696. The variation across folds reflects the natural difficulty of generalizing to different speaker groups: some groups of 13 speakers are acoustically closer to the training speakers than others.

VII. ERROR ANALYSIS

Looking at the confusion matrix, disgust is the class with the lowest recall (54.4%), and its most common error is being predicted as sad. This pattern makes sense acoustically: both disgust and sadness share low-energy, low-arousal prosody. In scripted speech, where the actor’s expression of disgust cannot rely on facial grimacing or body language, the vocal signal alone often sounds quiet and slightly flat exactly what sadness sounds like.

Fear is similarly difficult, frequently confused with anger at the high-arousal end of the spectrum and with sadness at

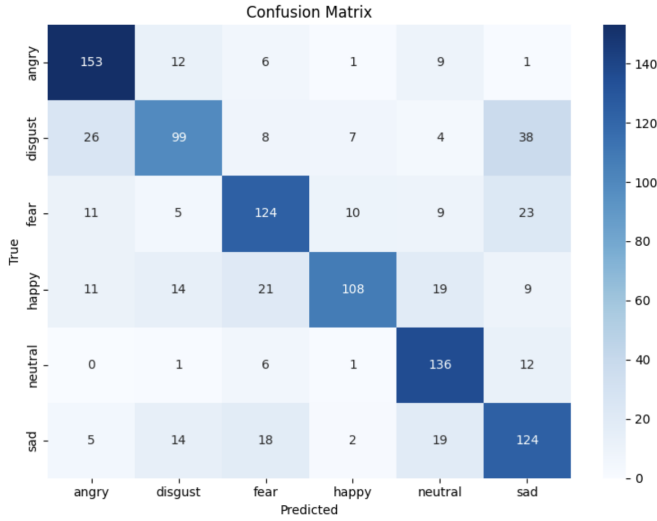


Fig. 4. Confusion matrix of the best multimodal fusion model on the test set

the low-arousal end. This reflects a well-known challenge in emotion recognition: emotions that are close to each other in valence-arousal space tend to share acoustic features, making them hard to separate without additional modality information.

Angry and neutral are the best-performing classes, with recalls of 84.5% and 87.1% respectively. Angry speech has a distinctive acoustic signature elevated energy, higher pitch, and faster speaking rate that Wav2Vec 2.0 learns to represent reliably. Neutral, by contrast, benefits from being a well-defined absence of emotional markers, which is perhaps easier to detect than specific positive or negative emotional expressions.

The error patterns also suggest that the text branch contributes meaningfully for some sentence-emotion pairings. Certain CREMA-D sentences appear predominantly in the training data with a particular emotion label not because the words are inherently emotional, but because the protocol assigned them to certain conditions more often. RoBERTa learns these distributional patterns, which can either help or mislead depending on whether test folds follow the same distribution. This is a limitation of using scripted speech for multimodal evaluation.

VIII. CONCLUSION

This paper presented a multimodal emotion recognition system that fuses speech representations from Wav2Vec 2.0 with text representations from RoBERTa using simple late-fusion concatenation. The system was trained and evaluated on CREMA-D under a strict speaker-independent five-fold cross-validation protocol.

Three key findings emerge. First, multimodal fusion outperforms both single-modality baselines by a substantial margin. Second, despite the text being informationally weak in the scripted CREMA-D setting, the text encoder still contributes useful signal, likely by learning distributional correlations between sentence identity and emotion label. Third, the biggest

bottleneck for audio-only performance is the pretrained backbone: wav2vec2-base, trained on read audiobooks, is not ideally suited to emotionally expressive speech.

The results suggest two productive directions for future work. First, replacing wav2vec2-base with a model pretrained on emotionally diverse or naturalistic speech such as wav2vec2-large-robust could close a meaningful performance gap on the audio side. Second, extending the architecture to include a visual modality (facial expressions) would complete the multimodal picture and is directly feasible given that CREMA-D includes video recordings for all utterances. With a properly trained visual branch, the full trimodal system could plausibly reach or exceed the 80.08% accuracy reported for audio-visual fusion on RAVDESS [6].

More broadly, this work demonstrates that simple fusion strategies concatenation of pretrained embeddings, selective fine-tuning of encoder layers, strict speaker-independent evaluation can achieve competitive results without architectural complexity.

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